Designing "Out of Eden": a simulation game blueprint

A prehistoric migration simulation covering 70,000 years can produce compelling emergent gameplay from surprisingly few rules. The core architecture needs just five interlocking layers — agents, environment, climate, culture, and a storyteller — each built on simple mechanics that compound into complex behavior. This report synthesizes population genetics, paleoclimate data, survival game design patterns, and simulation engineering into concrete, implementable formulas. The guiding principle throughout: every positive feedback loop must eventually trigger a negative one, and every catastrophe must feel like a chapter in an epic story rather than a random punishment.

Five layers that generate 70,000 years of emergent history

The minimum viable architecture draws from Sugarscape's resource competition, RimWorld's AI storyteller, Dwarf Fortress's cascade failures, and Civilization's geography-as-destiny. Each layer is deliberately simple; complexity emerges from their interactions.

Layer 1 — Agents (tribes/bands). Each tribe carries **6–8 attributes**: population count, food stores, genetic diversity metric, technology level, social cohesion, awareness range, tool quality, and migration state. Decision-making follows a weighted evaluation: assess local resources → compare to needs → if deficit, evaluate neighboring tiles → migrate or stay based on resource availability, social bonds, danger, and familiarity. This mirrors the Sugarscape architecture where 250 agents with vision, metabolism, and wealth on a 51×51 grid Wikipedia spontaneously produce migration waves, wealth inequality (Pareto distributions), tribal formation, and natural selection favoring good foragers. Academia.edu

Layer 2 — Environment (hex-grid tiles). Each tile provides **4 base variables**: food availability, water access, material resources, and danger level. Resources deplete when harvested and regenerate over time at biome-dependent rates. This depletion-regeneration cycle is the single most important driver of emergent migration — tribes naturally move when they've exhausted local resources, exactly mirroring the real Out-of-Africa dynamic.

Layer 3 — Climate (global + local). A pre-computed temperature curve drives biome shifts over millennia, while Perlin noise adds century-scale weather variation. Catastrophic events (volcanic winters, megadroughts) act as discrete modifiers. Details in the climate section below.

Layer 4 — Culture/knowledge. Technology bits diffuse between adjacent tribes like Sugarscape's cultural tags. Accumulation enables new resource exploitation (fishing, grinding seeds, bow hunting), creating the critical positive feedback loop: knowledge → better tools → more food → more people → more knowledge generators. This loop must eventually hit carrying capacity limits to prevent runaway growth.

Layer 5 — Storyteller. Borrowed from RimWorld's AI director, a "Nature" system manages event timing and intensity. It tracks tribal prosperity (food surplus, population trend, tool quality) and ensures growth periods alternate with crises. Game Wisdom Crucially, it uses **visible environmental storytelling** — drought, volcanic winter, ice advance — rather than invisible difficulty scaling. Players accept harsh mechanics when they're fictional forces of nature rather than arbitrary punishment. What Games Are

The design principle from emergence research: **"The best thing a designer can do is design a game that plays by itself without any other players, and still be interesting to watch."** Front Row Crew If the simulation generates compelling dynamics autonomously, player intervention makes it richer.

Feedback loops that drive the entire game

Every positive loop needs a natural negative counterpart:

- **Population growth → resource depletion → famine → population decline → resource recovery** (the core negative loop — prevents stagnation)
- **Knowledge accumulation → better tools → more food per person → population growth → carrying capacity exceeded** (positive → negative transition)
- **Trade networks → cultural exchange → technology diffusion → more trade goods → larger networks → attracts raiders/competition** (positive → negative)
- **Overhunting → prey scarcity → forced migration → prey recovery in abandoned area → eventual return** (spatial negative loop that drives migration)
- **Social cohesion → cooperative hunting → more food → more leisure → cultural development → stronger identity → resistance to change when adaptation needed** (hidden trap)

Calibrating population dynamics to real prehistory

Hunter-gatherer demographics follow remarkably consistent patterns across ethnographic studies. The numbers below come primarily from Gurven & Kaplan (2007), Dunbar (1992), Tallavaara et al. (2018), and the Binford ethnographic dataset.

Social organization scales in powers of three

Level	Size	Function	Game role
Intimate group	~5	Daily emotional support	Minimum survival unit

Level	Size	Function	Game role
Close friends	~15	Sympathy group	Hunting party size
Band	**25–50**	Co-resident foraging unit	**Core gameplay entity**
Community/clan	~150 (Dunbar's number)	Maximum meaningful relationships	Tribe identity threshold
Mega-band	~500–600	Seasonal gatherings	Regional alliance
Tribe	~1,500–2,000	Linguistic/cultural group	Cultural boundary

Each layer is roughly **3× the one inside it.** PubMed Central A band below ~25 people cannot sustain itself demographically. For the game, the band of 25–50 is the fundamental unit; ScienceDirect Dunbar's number of ~150 is the threshold where a tribe must split or develop hierarchy. Wikipedia

Growth, death, and the forager population paradox

Long-term archaeological growth rate: **~0.04% per year** (doubling time ~1,700 years). PNAS But ethnographic hunter-gatherers show short-term growth of **1–3% per year.** PubMed Central The resolution: periodic catastrophic crashes (famine, disease, drought, conflict) reduce the long-term average to near zero. PNAS +2 **For the game, model a potential growth rate of 0.5–1.5% under favorable conditions, with periodic crashes that bring the long-run average to ~0.04%.** This creates the dramatic boom-bust cycles that make survival games compelling.

Life expectancy at birth was **21–37 years** (mean ~30), Cambridge Core but this is misleading. Ucsb **Infant mortality ran ~27%** and child mortality before age 15 was **~49%** Cold Takes — roughly half of all children died before adulthood. Paleo Leap Those surviving to age 15 could expect to live to **~54**, Wikipedia and the modal adult death age was **68–78 years.** Paleo Leap The Paleo Diet The game should model childhood as the dangerous period, not adulthood. Primary cause of death: infectious disease, followed by violence (~8%), accidents, and starvation. Cambridge Core

Carrying capacity by biome

Biome	Density (persons/km²)	Per band of 40
Rich coastal/estuarine	0.5–2.0+ (up to 8.4)	20–80 km²
Temperate forest	0.1–1.15	35–400 km²
Tropical savanna	0.1–0.5	80–400 km²
Steppe/grassland	0.02–0.15	270–2,000 km²

Biome	Density (persons/km²)	Per band of 40
Boreal forest	0.01–0.05	800–4,000 km²
Desert	0.01–0.15	270–4,000 km²
Tundra/Arctic	0.003–0.02	2,000–13,000 km²

Coastal and riverine zones are 3–10× more productive than their inland equivalents (PNAS) — this drives the coastal migration hypothesis for the Out-of-Africa expansion and should be a major gameplay incentive. Megafauna presence multiplies carrying capacity by ~1.5×, making their extinction a devastating mid-game event.

Global population trajectory (the game's population target curve)

Date	Global population	Key context
70,000 BC	~100,000–300,000	MIS 4 cold phase; Africa only
50,000 BC	~200,000–500,000	Out-of-Africa expansion underway
30,000 BC	~1–5 million	Upper Paleolithic; Europe colonized
15,000 BC	~2–5 million	End of Last Glacial Maximum
10,000 BC	~4–10 million	Early Holocene; pre-agriculture

Migration frontier advanced at **~0.5–1.5 km/year**, (Wikipedia) with ~140 km between successive colonization events every ~280 years (~10 generations). (PNAS) Africa to Australia (~25,000 km) took ~10,000–20,000 years along the coastal route. (Wikipedia)

Engineering a 10% extinction probability

The game needs total human extinction in roughly 1 out of 10 runs. Pure independence between tribes makes this nearly impossible — with 5 independent tribes each needing a **63% individual extinction rate** to produce 10% total extinction, individual games would be miserable. The solution is **correlated catastrophes**.

The correlated catastrophe architecture

```
P(total_extinction) = P(catastrophe) × P(all_die|catastrophe) + P(no_catastrophe)
```

```
× P(all_die|no_catastrophe)
```

Recommended calibration:

- P(global catastrophe per run) = **0.15** (Toba-scale volcanic winter, megadrought)
- P(all tribes die | catastrophe) = **0.50** (depends on preparation and luck)
- P(all tribes die | no catastrophe) = **0.02** (from accumulated independent bad luck)
- **Total: 0.15 × 0.50 + 0.85 × 0.02 = 0.092 ≈ 10%**

This means most runs (~85%) face no civilization-ending catastrophe and have only a 2% extinction risk from accumulated misfortune. But when a supervolcano erupts or a megadrought strikes, it's a coin flip. Players who prepared (diversified geographically, built food stores, developed fishing technology) survive more often. This makes extinction feel **earned** rather than random.

Allee effects create the extinction vortex

Below a critical population threshold, per-capita growth rate turns **negative** — the strong Allee effect. (nature) (Wikipedia) Model this with the cubic population equation:

```
dN/dt = r × N × (N/A - 1) × (1 - N/K)
```

Where **A** is the Allee threshold (~20–80 individuals), **K** is carrying capacity, and **r** is growth rate. Three equilibria emerge: extinction (stable), the threshold A (unstable tipping point), and K (stable). (Nature) If population drops below A, it spirals to zero. This creates dramatic near-miss scenarios where a tribe hovers just above the threshold.

The **extinction vortex** (Gilpin & Soulé 1986) compounds through four reinforcing loops: demographic instability at low numbers, spatial fragmentation, genetic drift causing inbreeding depression, and loss of adaptive potential. (Wikipedia) (bioRxiv) For the game, model inbreeding as:

```
F(t+1) = F(t) + (1 - F(t)) / (2 × Ne)
fitness_penalty = δ × F(t)    [δ = 3–12 "lethal equivalents"]
```

Where **Ne** (effective population) is roughly **0.6–0.7 × census population** for hunter-gatherers. (PubMed Central) When Ne drops below ~50–100, inbreeding depression becomes measurable within 5 generations. The revised conservation genetics consensus (Frankham 2014) recommends **Ne ⩾ 100** for short-term viability (ScienceDirect) and **Ne ⩾ 1,000** for long-term evolutionary potential (ScienceDirect) — translating to census populations of ~150 and ~1,500 respectively.

Tunable extinction parameters

Parameter	Range	Effect
Allee threshold (A)	20–80 individuals	Higher = more extinction risk
Environmental variance (σ^2)	0.01–0.5	Higher = more stochastic extinction
Catastrophe frequency	0.001–0.05 per generation	Higher = more extinction risk
Inter-tribe correlation (ρ)	0–1	Higher = more total-extinction risk
Migration rate between tribes	0–0.1 per generation	Higher = less extinction risk (genetic rescue)
Number of independent tribes	3–10	More = less total-extinction risk

The single most sensitive parameter is inter-tribe correlation (ρ). When a climate catastrophe hits all tribes simultaneously, geographic diversification provides no safety net. This is historically accurate — correlated environmental shocks (volcanic winters, Heinrich events) are what actually threatened human survival.

Making failure feel earned, not arbitrary

The research on Dwarf Fortress, Frostpunk, Against the Storm, and Banished reveals a consistent framework for "hard but fair" design.

Input randomness over output randomness

This distinction is the most important principle. **Input randomness** (what situation you face) feels fair — "the environment was harsh but I chose how to respond." **Output randomness** (whether your action succeeds) feels unfair — "I made the right choice but the dice said no." For Out of Eden, randomize the environment (terrain, climate events, animal migrations, disease outbreaks) but make player decisions about preparation, migration, and resource allocation largely deterministic in their effects. When hunters fail, the player should see **why** — wrong season, inadequate tools, depleted hunting grounds — not just "hunt failed, roll 1."

Four-tier crisis escalation

Best survival game design uses a gradient of crisis severity with intervention points at each

stage:

1. **Minor crisis** — easily fixable with attention (one bad winter, temporary food shortage)
2. **Moderate crisis** — fixable with sacrifice (abandon a project, slaughter breeding animals, reduce tribe size)
3. **Severe crisis** — fixable with drastic action (split the tribe, abandon territory, merge with another group)
4. **Terminal** — only reached through sustained neglect or truly overwhelming catastrophe

Pure death spirals (where each loss accelerates the next with no exit) are the most frustrating pattern in survival games. StrateGGames Banished's food-production spiral is the canonical example. Steam Community **Every death spiral needs an escape valve** — emergency foraging, sacrifice of stored goods, temporary migration, asking neighboring tribes for aid. Blogger The spiral should be **diagnosable** so players understand why they're failing.

The Frostpunk principle: survival decisions as cultural identity

Frostpunk's Book of Laws system Gabrielchauri demonstrates that the most meaningful choices in survival games aren't resource optimizations — they're **moral dilemmas that define who your people become**. For Out of Eden, consider irreversible cultural decisions during crises: Do you practice infanticide during famine? Abandon the elderly on migration? Share equally or prioritize hunters? Adopt the customs of a more successful neighboring tribe? These decisions should have lasting mechanical AND narrative consequences, creating divergent tribal identities across playthroughs. The formula: **resource scarcity** + **competing values** + **irreversibility** = **meaningful choice**.

The 65% rule and hidden pity counters

Research cited by game designers suggests players need to experience victories at least **65% of the time** to perceive a game as fair. Z League After consecutive bad outcomes (three harsh winters in a row, multiple failed hunts), implement a hidden "pity counter" that slightly increases the probability of favorable events. Players over-remember failures and dismiss successes Z League — the system must compensate for this negativity bias without being detectable.

A computationally cheap climate engine

The game's 70,000-year timeframe demands a climate model that's simple to compute but produces genuinely interesting variation. The approach: a pre-computed global temperature curve with noise layers, driving a Whittaker biome lookup per tile.

Global temperature as a lookup table with noise

```
T_global(year) = interpolate(CLIMATE_CURVE, year)    // ~15 data points
               + DO_amplitude × sin(year/1500 × 2π)  // Dansgaard-Oeschger
oscillation
               + 0.5 × perlinNoise(year/200)          // century-scale weather
```

Key data points for the curve: MIS 4 at −4°C (70,000 BP), MIS 3 at −2.5 to −3°C (57,000–29,000 BP), **Last Glacial Maximum at −6°C** (26,500–19,000 BP), Wikipedia Bølling-Allerød at −2°C (14,700 BP), Wikipedia Younger Dryas at −4°C (12,900 BP), Holocene at 0°C (11,600 BP onward). Encyclopedia Britannica The D-O oscillation amplitude should be **~2°C during glacial periods, ~0.3°C during the Holocene**. These oscillations, occurring roughly every 1,500 years, Simple Book Publishing are the most dramatic gameplay-relevant rapid events — a D-O warming can transform tundra into boreal forest within a generation. Wikipedia

Per-tile temperature and biome assignment

```
T_tile = (30 - 0.7 × |latitude|) + glacial_offset × (1 + 1.5 × (|lat|/90)²)
         - 6.5 × altitude_km + coastal_modifier
```

Feed temperature and precipitation into a simplified **Whittaker biome lookup** — a 2D array indexed by (mean annual temperature, annual precipitation) returning a biome enum. When the global offset shifts by >0.5°C, recompute all tile biomes. Biomes naturally migrate equatorward during cooling and poleward during warming.

Sea level drives the most dramatic map changes

Sea level at the LGM sat **120–130m below present**, ScienceInsights Wikipedia exposing Beringia (connecting Asia to the Americas, up to 1,600 km wide), Encyclopedia Britannica Sundaland (connecting Borneo, Java, Sumatra to mainland Asia, 1.85 million km² of land), Wikipedia and Doggerland (connecting Britain to Europe). Implementation is a simple threshold: `tile_exposed = tile.elevation > sea_level(year)`. Meltwater Pulse 1A at ~14,700 BP raised sea levels **~20m in 500 years** Nature Wikipedia — fast enough to flood coastal tiles within a few generations, forcing dramatic migrations and destroying established settlements.

Megafauna extinction as a mid-game resource crisis

88% of Australian megafauna vanished by 40,000 BP; **72% of North American megafauna** disappeared between 13,000 and 11,000 BP. Wikipedia Megafauna provide a large carrying capacity bonus (high-calorie hunting with cooperative techniques). Their extinction — driven by a combination of human overkill and climate change Encyclopedia Britannica — should be the game's most consequential mid-to-late event, forcing the dietary diversification that historically led to the Broad Spectrum Revolution and eventually agriculture.

From foraging to farming: the subsistence technology tree

The 70,000-year span covers a profound transformation in how humans obtained food. Rather than a discrete "discover agriculture" event, the real transition involved centuries of intermediate stages that map naturally to a technology progression.

Stage 1 — Pure hunting and gathering (70,000–23,000 BP). Diet varies by latitude: tropical environments provide ~50–80% plant foods, high-latitude environments are ~80–100% animal-based. A band of 25 forages **3–5 hours per day** per adult (Rewild) (PubMed Central) and needs ~50,000–65,000 kcal daily. Key technologies: controlled fire, hafted spears, projectile weapons (atlatl ~70,000–40,000 BP), early bone harpoons for fishing.

Stage 2 — The Broad Spectrum Revolution (23,000–10,000 BP). Triggered by population pressure in optimal habitats, megafauna decline, and post-LGM climate shifts. (ScienceDirect) Diets diversify from reliance on large game to include small game, fish, shellfish, seeds, and nuts. (Wikipedia) Key technologies: **grinding stones** (for processing seeds into flour, ~23,000 BP), microliths (composite tools), nets/snares/traps, (PNAS) fish hooks (oldest at ~42,000 BP in East Timor), (Grokipedia) and **storage pits**. This stage unlocks new carrying capacity from previously ignored resources.

Stage 3 — Semi-sedentism and intensive harvesting (15,000–11,500 BP). The **Natufian culture** in the Levant exemplifies this: permanent stone dwellings, sickle blades for wild cereal harvesting, elaborate mortuary practices, and proto-domestication of plants and dogs. (Wikipedia) (Dalvoy) Natufians derived up to ~70% of their diet from wild grains. (Dalvoy) The house mouse self-domesticated around grain stores. (Academia.edu) This stage requires prior technologies (grinding stones, sickles, storage) and favorable environmental conditions (dense wild cereal stands).

Stage 4 — Proto-agriculture and domestication (11,500–8,000 BP). The Younger Dryas (12,900–11,600 BP) (Encyclopedia Britannica) may have been the final trigger: climate stress forced semi-sedentary Natufians to actively cultivate rather than merely harvest. (Wikipedia) Full morphological domestication (non-shattering grain rachis) took **~1,000–3,000 years** of selection pressure. Dogs appeared in Natufian burials by ~12,000 BP; (Grokipedia) sheep were domesticated by ~11,000 BP, goats by ~10,000 BP.

For game design, each stage should unlock through a combination of accumulated technology points AND environmental conditions. The Broad Spectrum Revolution shouldn't fire just because a tribe researches it — it should emerge when megafauna decline and population pressure force dietary diversification. Agriculture shouldn't be "discovered" — it should become the rational response when semi-sedentary tribes with grain-processing technology face climate stress.

Engineering a simulation that runs 70,000 ticks cleanly

A simulation spanning 70,000 annual ticks with potentially thousands of tribe entities demands careful architectural decisions. The key recommendations: **ECS architecture with Structure-of-Arrays data layout, integer arithmetic for all countable quantities, priority queues for rare events, and multi-rate time stepping.**

Architecture and data layout

Use an **Entity Component System** where tribes are entity IDs associated with component arrays (population, position, food stores, tech level, genetic diversity). Store these as Structure of Arrays — all population values contiguous in memory, all positions contiguous, etc. For 5,000 tribes, the "hot" data (population + position + food stores) totals ~60KB, fitting entirely in L2 cache. This enables SIMD vectorization and eliminates cache misses on the main simulation loop.

Use **hex grids with cube coordinates** for spatial logic. Distance calculation is trivial: `max(|Δq|, |Δr|, |Δs|)`. (The Knights of U) Each tile maintains a list of resident tribe IDs, providing O(1) lookup for local interactions without any complex spatial data structure. For 2,000–3,000 tiles covering the Old World and Americas (~500–1,000 km per tile), A* pathfinding (GitHub) explores at most a few hundred nodes. **Dijkstra flow fields** are even better: when many tribes migrate toward the same region, compute one distance map and let all tribes follow the gradient.

Numerical stability over 70,000 ticks

Use integers for all countable quantities. Population counts, resource amounts, and technology levels should never be floating-point. A 32-bit signed integer holds ~2.1 billion, more than sufficient for any individual tribe. Use 64-bit integers for intermediate multiplication results.

For the few quantities requiring fractional math (genetic diversity, growth rates), use **fixed-point Q32.16 arithmetic** — 32 integer bits and 16 fractional bits. This provides exact cross-platform determinism with no NaN, no infinity, no denormalized-number surprises. The Photon Quantum game engine uses Q48.16 fixed-point for all multiplayer simulation.

For the random number generator, use **PCG64 or xoshiro256++** with explicit seeded state stored as part of the simulation. Over 70,000 ticks with 5,000 tribes using ~10 random numbers per tribe per tick, total consumption is ~3.5 billion numbers — well within PCG64's period of 2^{128}.

Multi-rate updates and event scheduling

Not every system needs updating every tick:

Subsystem	Frequency	Rationale
Population growth	Every tick	Core mechanic
Resource consumption	Every tick	Tied to population
Migration decisions	Every 4 ticks	Tribes don't relocate yearly
Climate/biome recalculation	Every 10–50 ticks	Climate changes slowly
Technology advancement	Every 10 ticks	Discoveries are rare
Genetic drift/inbreeding	Every 25–50 ticks	Very slow process
Inter-tribal interaction	Event-driven	Only when tribes share a tile

For rare events (volcanic eruptions, technological breakthroughs, epidemics), **never check every tribe every tick**. Instead, pre-schedule using the geometric distribution: `ticks_until_next = floor(ln(1-rand()) / ln(1-probability))` and store in a min-heap priority queue. This converts O(numTribes) per-tick checks into O(numEvents) processing.

Memory management and history

Use **object pools** with generational indices for tribe entities. Game Programming Patterns Pre-allocate capacity for ~10,000 tribes at startup; recycle slots when tribes die. For history logging, use **logarithmic sampling**: keep every data point for the last 100 ticks, every 10th point for ticks 100–1,000 ago, every 100th beyond that. This gives ~340 data points for 70,000 ticks instead of 70,000. Full checkpoints every 1,000 ticks enable time-travel debugging via binary search. Rescale

Preferred language: Rust (zero-cost abstractions, no GC, memory safety, excellent ECS libraries like bevy_ecs and hecs). C++ is the strong alternative. Avoid Python for the core simulation loop — reserve it for prototyping only.

The complete per-tick population formula

Bringing everything together into a single implementable model:

```
N(t+1) = N(t) × exp[r_eff × (1 - N(t)/K(t)) × max(0, (N(t) - A)/N(t))] + migration
+ noise

Where:
  r_eff = r_base - inbreeding_penalty(F) - environmental_stress(t)
  K(t)  = base_biome_capacity × coastal_bonus × megafauna_bonus ×
catastrophe_modifier × tech_multiplier
  F(t+1) = F(t) + (1 - F(t)) / (2 × Ne)
  inbreeding_penalty = δ × F(t)     [δ ≈ 6 lethal equivalents]
  noise ~ Normal(0, σ_d²/N(t) + σ_e²)  [demographic + environmental stochasticity]
```

The `(N(t) - A) / N(t)` term implements the Allee effect: when population drops below threshold A, growth turns negative and the tribe spirals toward extinction. Royal Society Open Science The `(1 - N(t)/K(t))` term implements logistic carrying capacity. Mathematics LibreTexts The inbreeding coefficient F accumulates over generations, creating the genetic extinction vortex. Sustainability-directory Environmental stochasticity (σ_e^2) affects all tribes in a region simultaneously, Wikipedia while demographic stochasticity (σ_d^2/N) hits small tribes harder Cell Press Springer — exactly matching ecological reality.

Conclusion

The most actionable insight across all seven research domains is that **emergence comes from coupling, not complexity**. Five simple layers Technovation — agents with needs, a depletable environment, a shifting climate, diffusing knowledge, and a storyteller pacing events — will produce 70,000 years of migration, tribal formation, cultural divergence, and near-extinction drama without scripting any of it. The correlated-catastrophe architecture solves the 10% extinction requirement elegantly: most runs are survivable through good play, but supervolcanic winters create genuine existential crises where preparation determines survival. The discrete logistic growth equation with an Allee threshold is the single formula that captures both boom-bust population dynamics and extinction-vortex spirals. Royal Society Open Science And the Broad Spectrum Revolution — humanity's real shift from megafauna hunting to diversified foraging to proto-agriculture — provides a natural technology progression that emerges from environmental pressure rather than arbitrary research trees. Build these systems with ECS architecture, integer arithmetic, and multi-rate time stepping in Rust, and the simulation will run 70,000 ticks in seconds without drift, leaks, or overflow.